DO NOW — RECALL

Lesson 104 · Magnetic, Optical and Flash Storage

Name: _____ Class: _____ Date: _____

1. What does non-volatile mean?

2. What is seek time?

3. What does optical storage use to read data?

4. 1 byte contains how many bits?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Retains contents without power.

2. Model answer

Time to position/locate data for a storage access.

3. Model answer

Light/laser.

4. Model answer

8 bits.

By the end, you can…

• Magnetic hard disks use magnetised regions on spinning platters with moving heads; they can offer large capacity at relatively low cost per GB.

• Optical discs are read with a laser, often used for physically distributed or archived/removable media, though capacity and drive availability vary.

• Flash uses electronic non-volatile memory cells without moving read/write parts; often faster random access and more robust to vibration than HDDs, but trade-offs remain.

Vocabulary — know these terms

magnetic — Storage via magnetisation.

optical — Storage read with laser/light.

flash — Non-volatile electronic solid-state storage.

cost per GB — Price relative to capacity.

NEW LEARNING

Read, annotate and explain

Understand

Magnetic hard disks use magnetised regions on spinning platters with moving heads; they can offer large capacity at relatively low cost per GB.

Apply

Optical discs are read with a laser, often used for physically distributed or archived/removable media, though capacity and drive availability vary.

Evaluate

Flash uses electronic non-volatile memory cells without moving read/write parts; often faster random access and more robust to vibration than HDDs, but trade-offs remain.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Choose portable storage for a field photographer travelling over rough ground with active projects.

Explain the choice

Flash SSD: no spinning platter and generally quick random access; still requires backups.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Match HDD, DVD and USB SSD to magnetic, optical and flash storage, with mechanisms. [6]

Question 2

Explain one advantage and one limitation of flash SSD over magnetic HDD for a mobile video editor. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Evaluate storage choice for distributing 100 offline revision packs to pupils and for a multi-terabyte editable school video archive. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A hard disk drive reads bits using a laser.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.